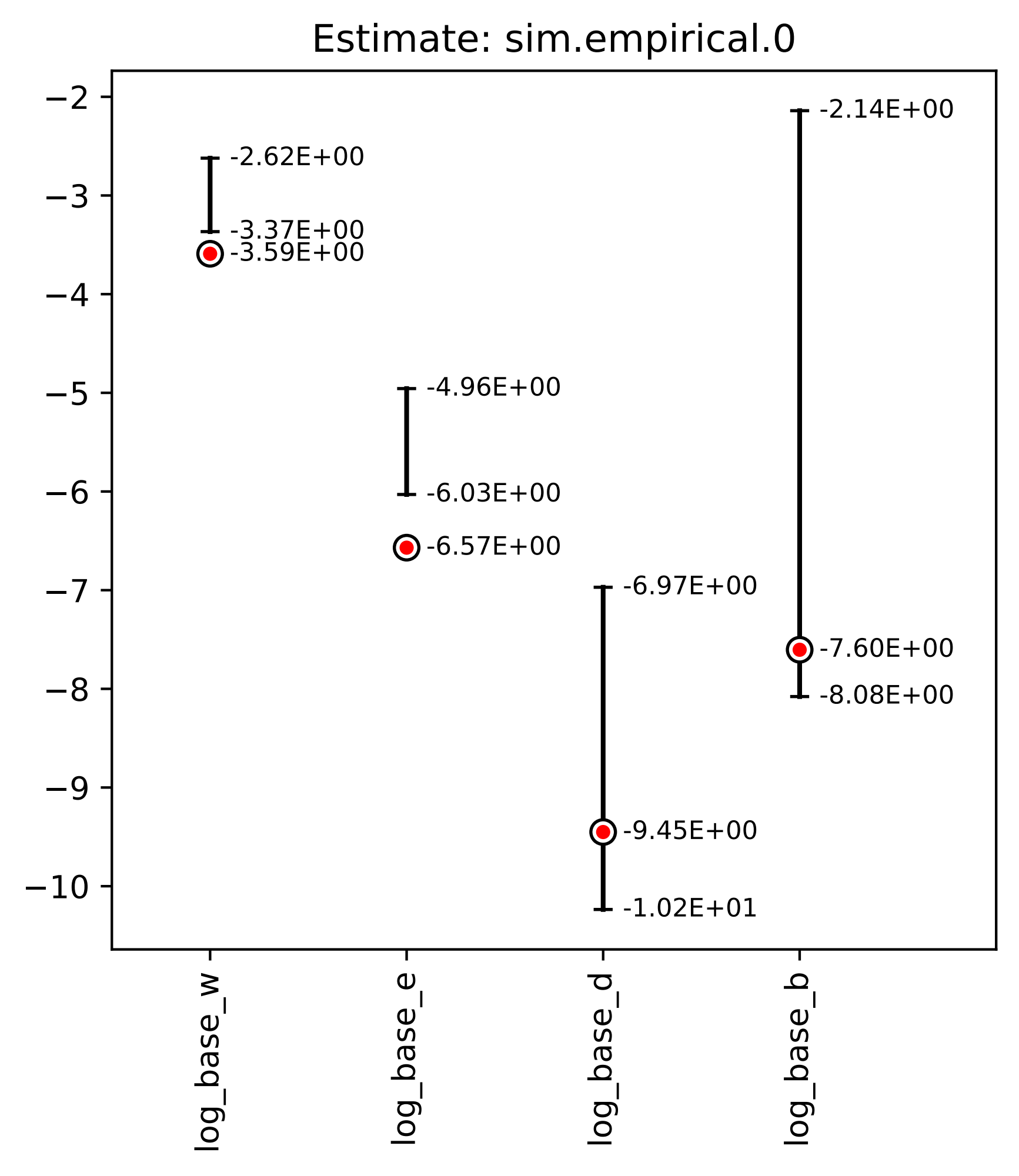
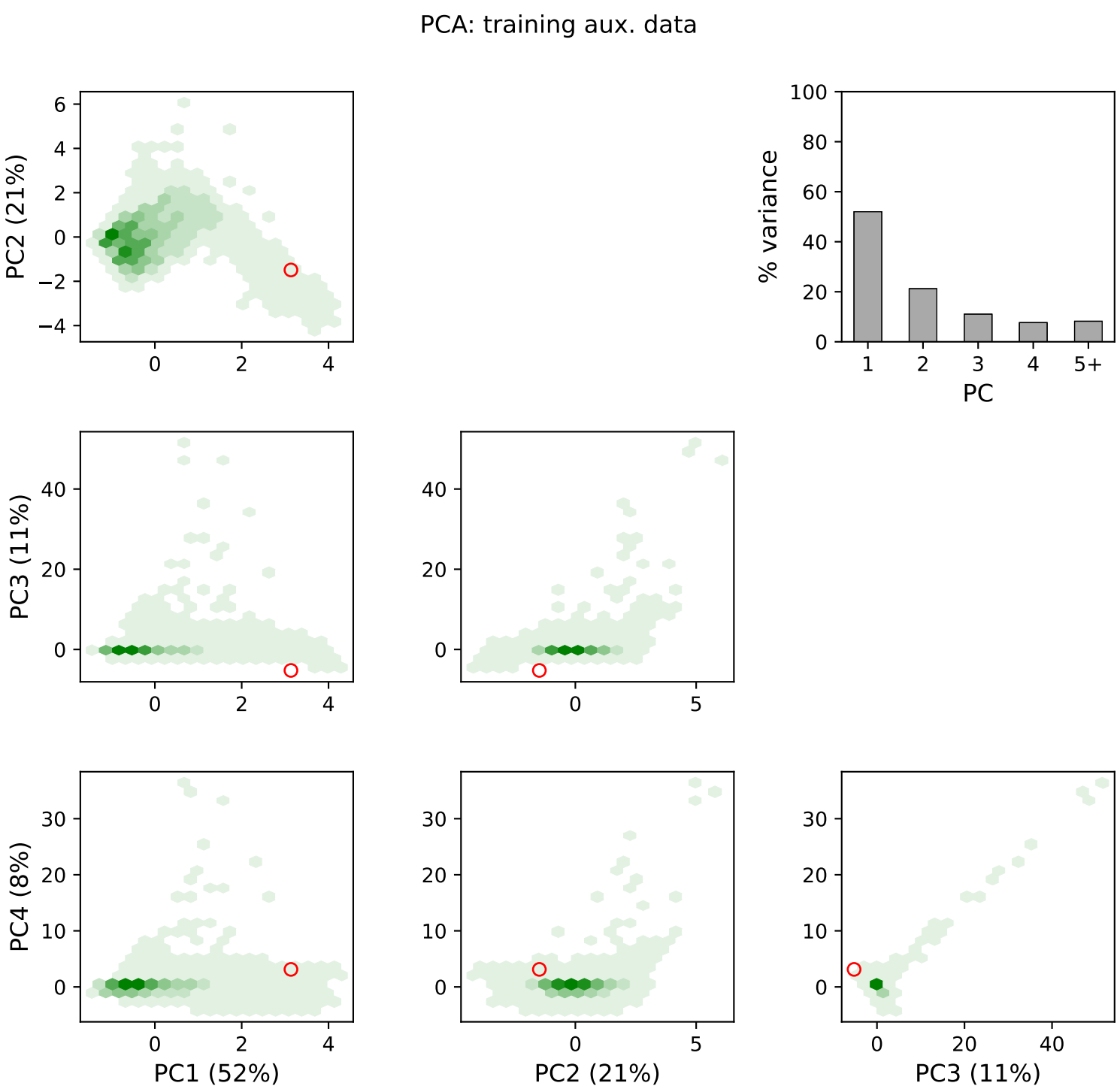
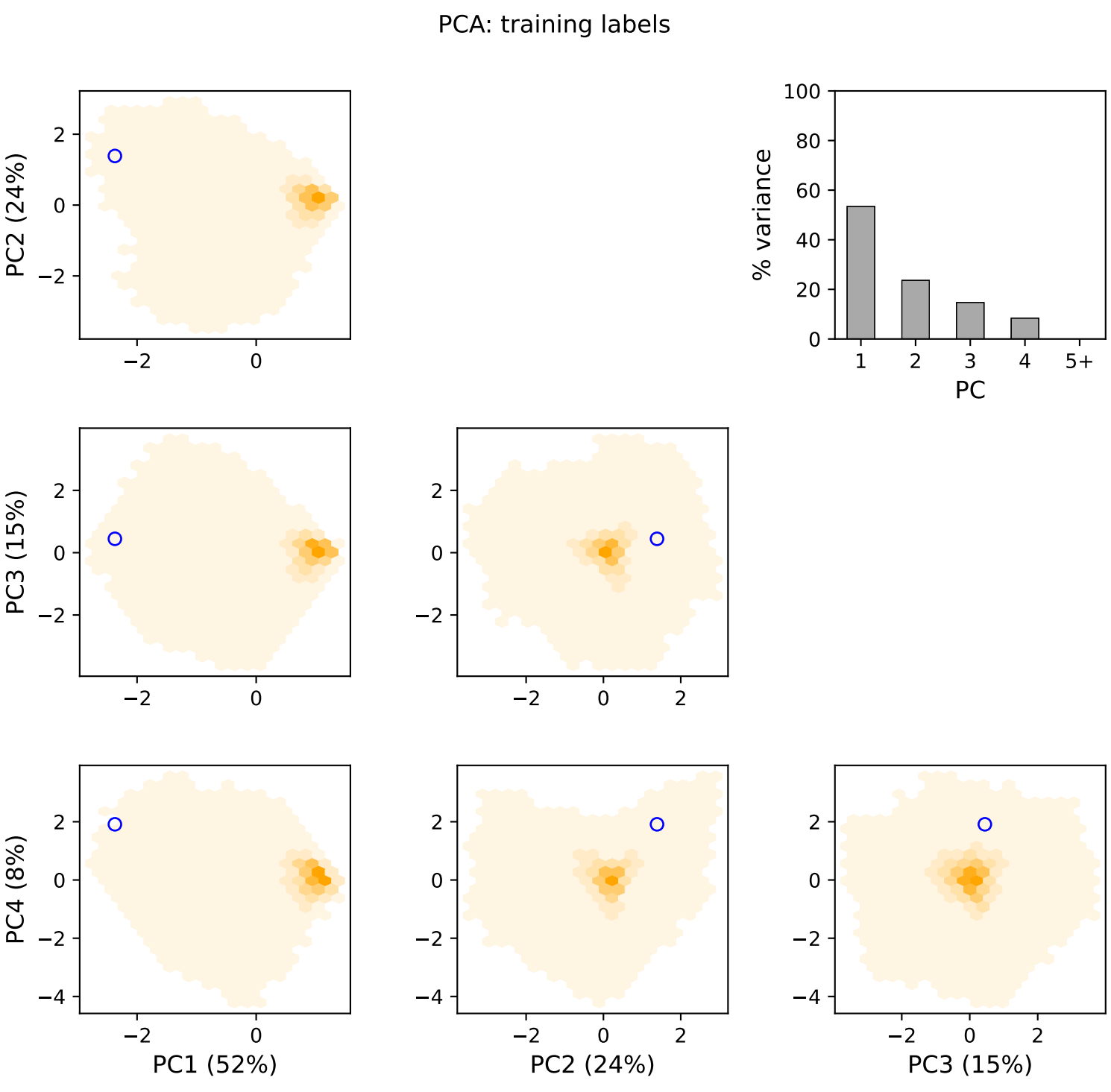
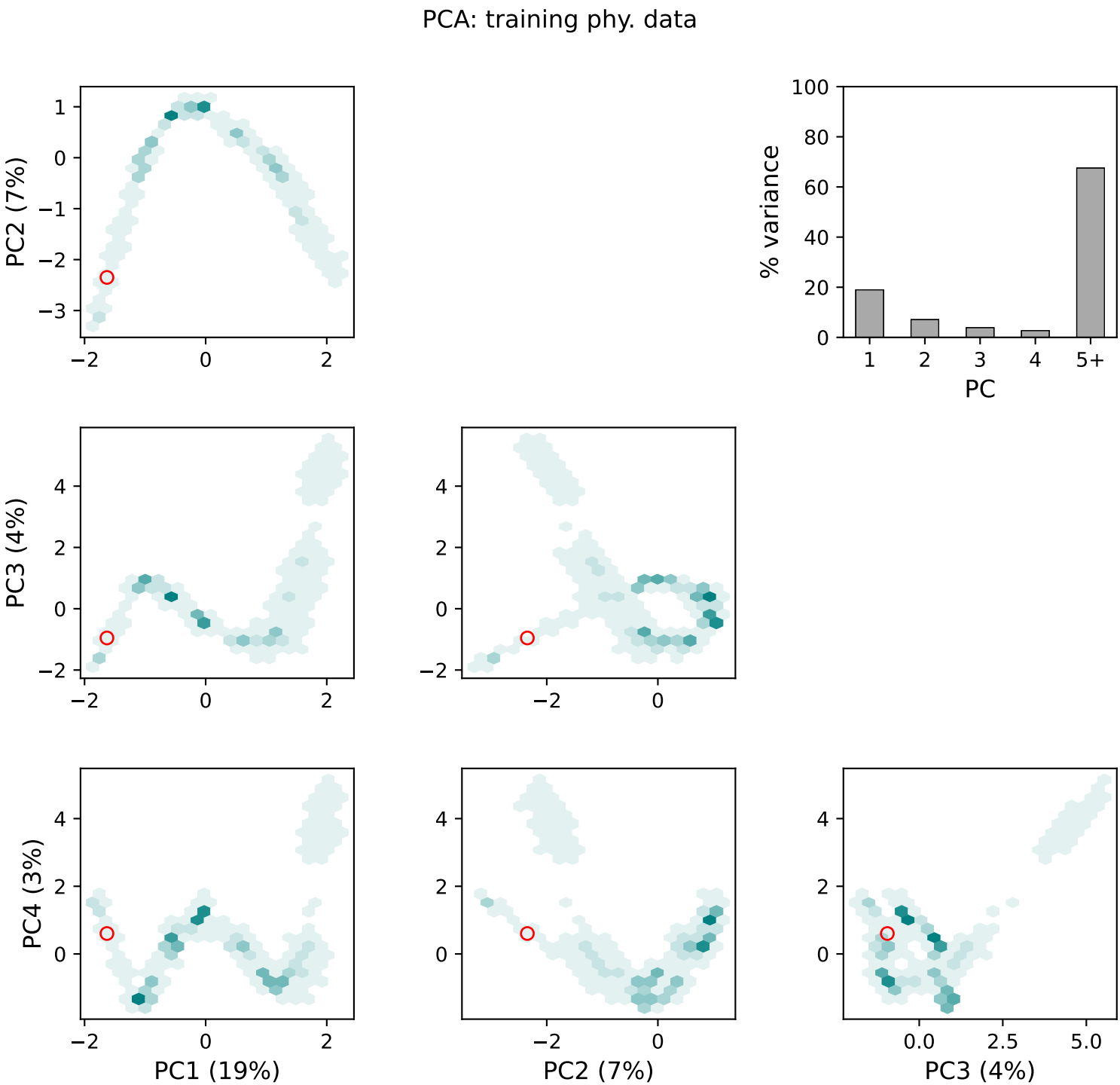
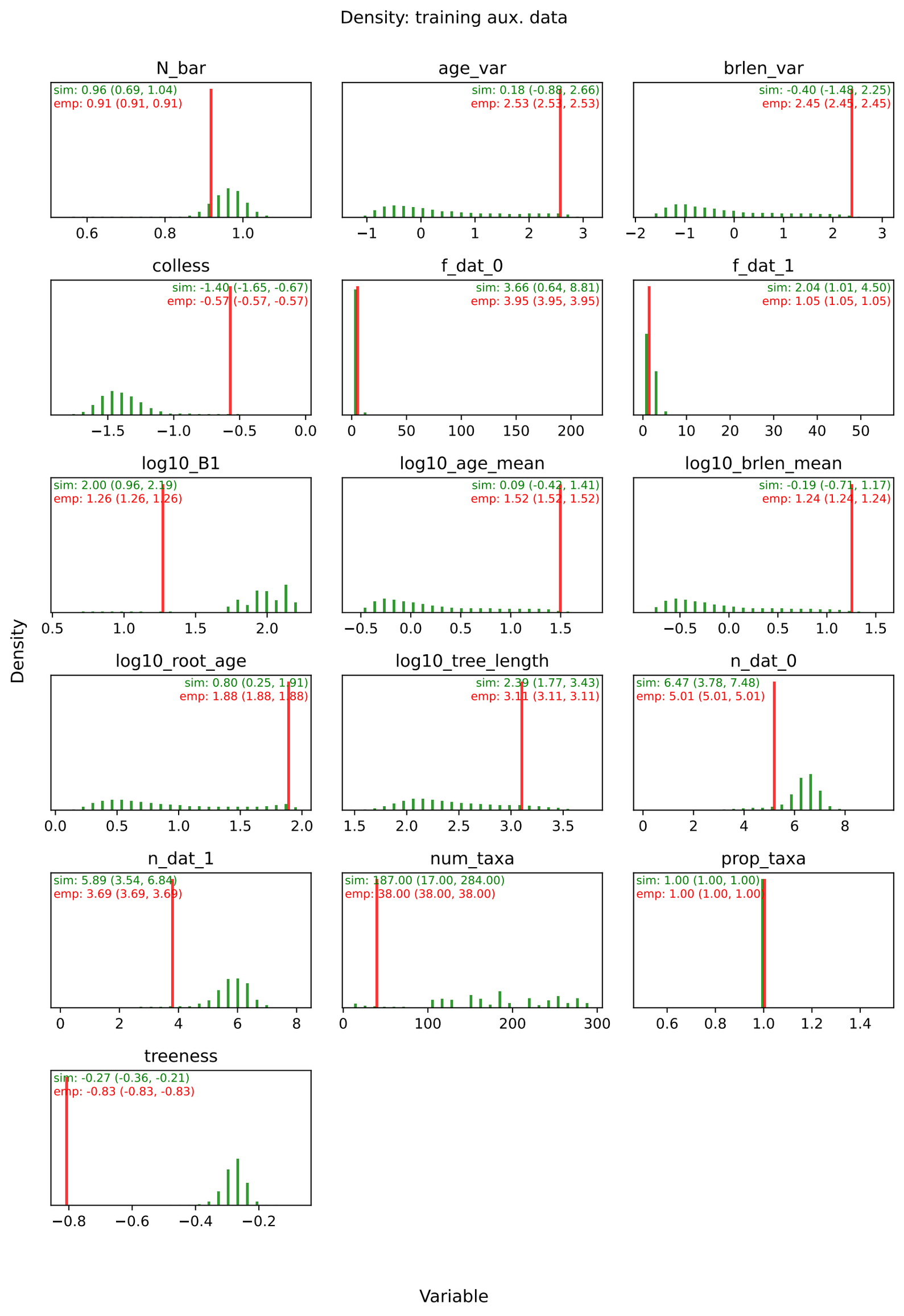
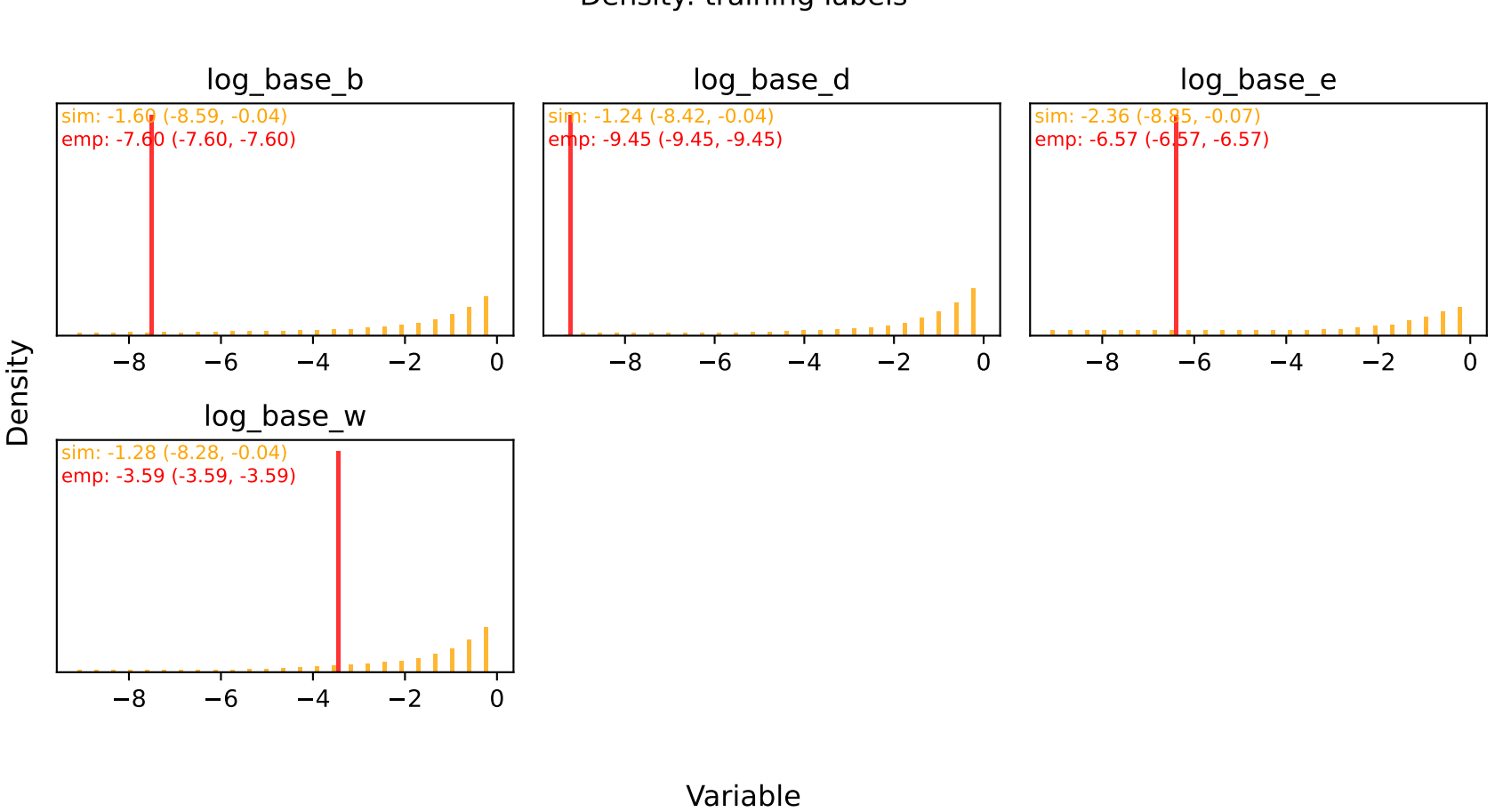

Estimate: sim.empirical.0
−2
−3
−4
−5
−6
−7
−8
−9
−10
-2.62E+00
-3.37E+00
-3.59E+00
-4.96E+00
-6.03E+00
-6.57E+00
-6.97E+00
-9.45E+00
-1.02E+01
-2.14E+00
-7.60E+00
-8.08E+00
log_base_w
log_base_e
log_base_d
log_base_b

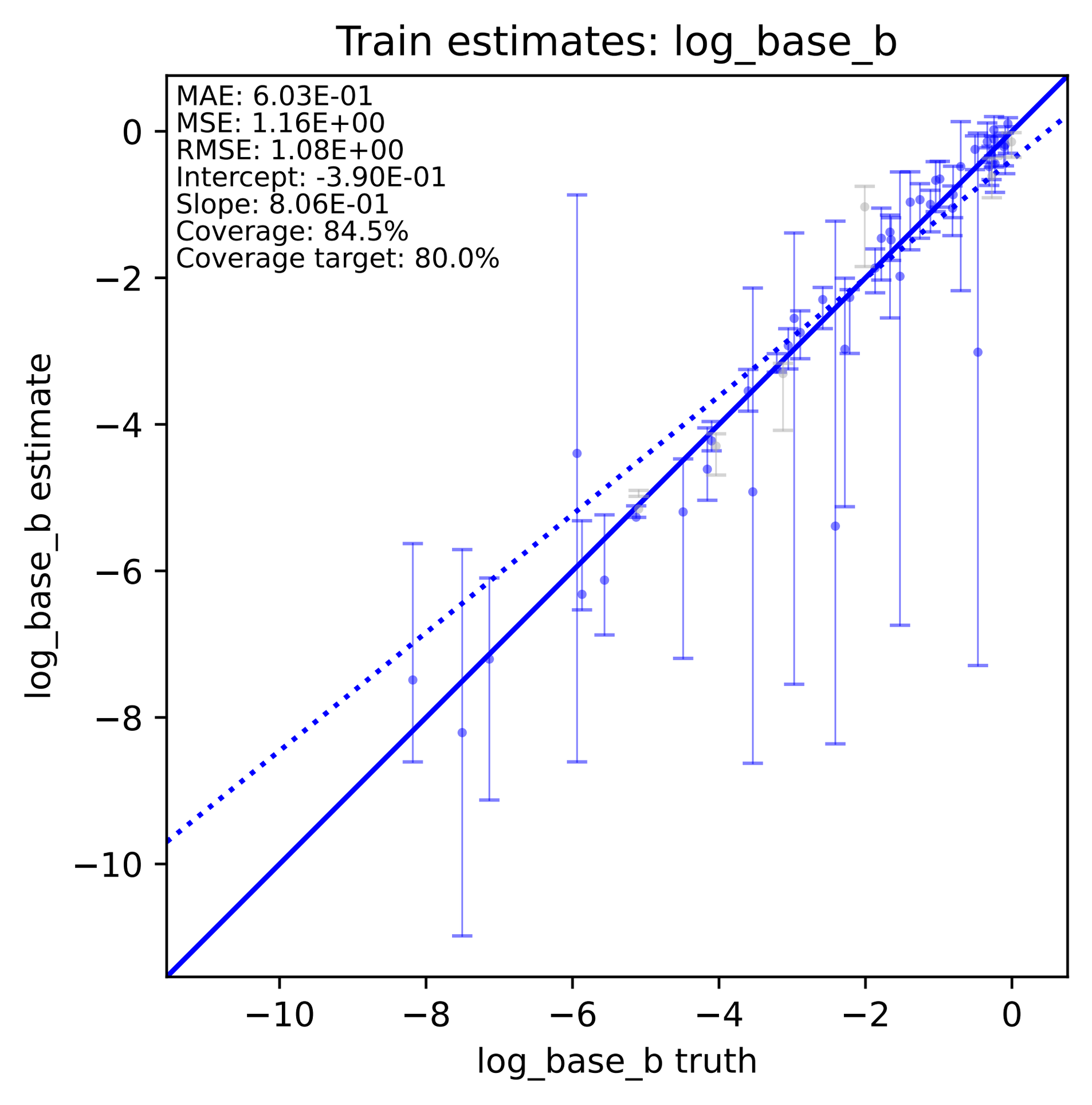

PCA: training aux. data
PC2 (21%)
% variance
PC
PC3 (11%)
PC4 (8%)
PC1 (52%)
PC2 (21%)
PC3 (11%)

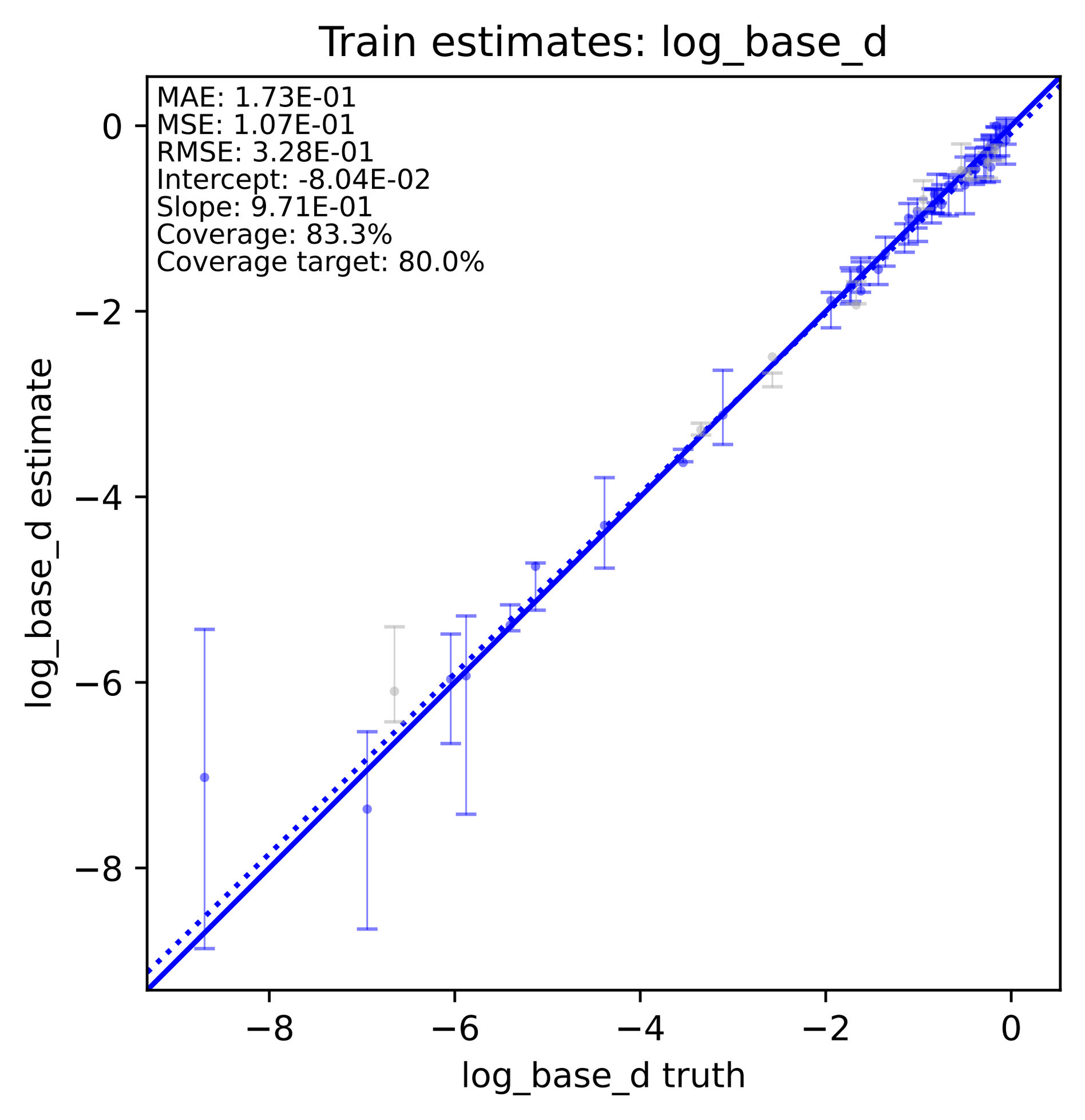

PCA: training labels
PC2 (24%)
% variance
PC
PC3 (15%)
PC4 (8%)
PC1 (52%)
PC2 (24%)
PC3 (15%)

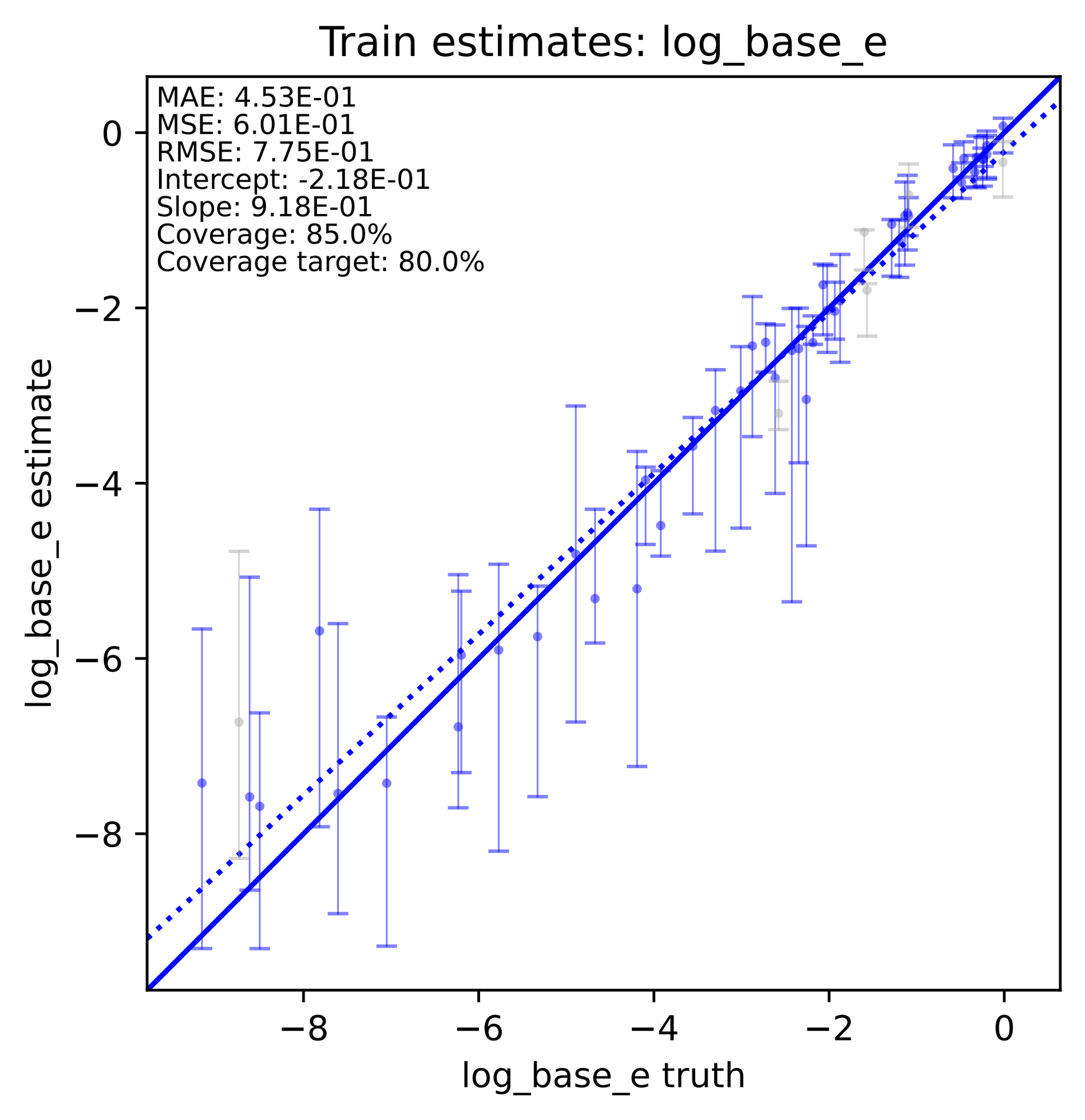

PCA: training phy. data
PC2 (7%)
% variance
PC
PC3 (4%)
PC4 (3%)
PC1 (19%)
PC2 (7%)
PC3 (4%)

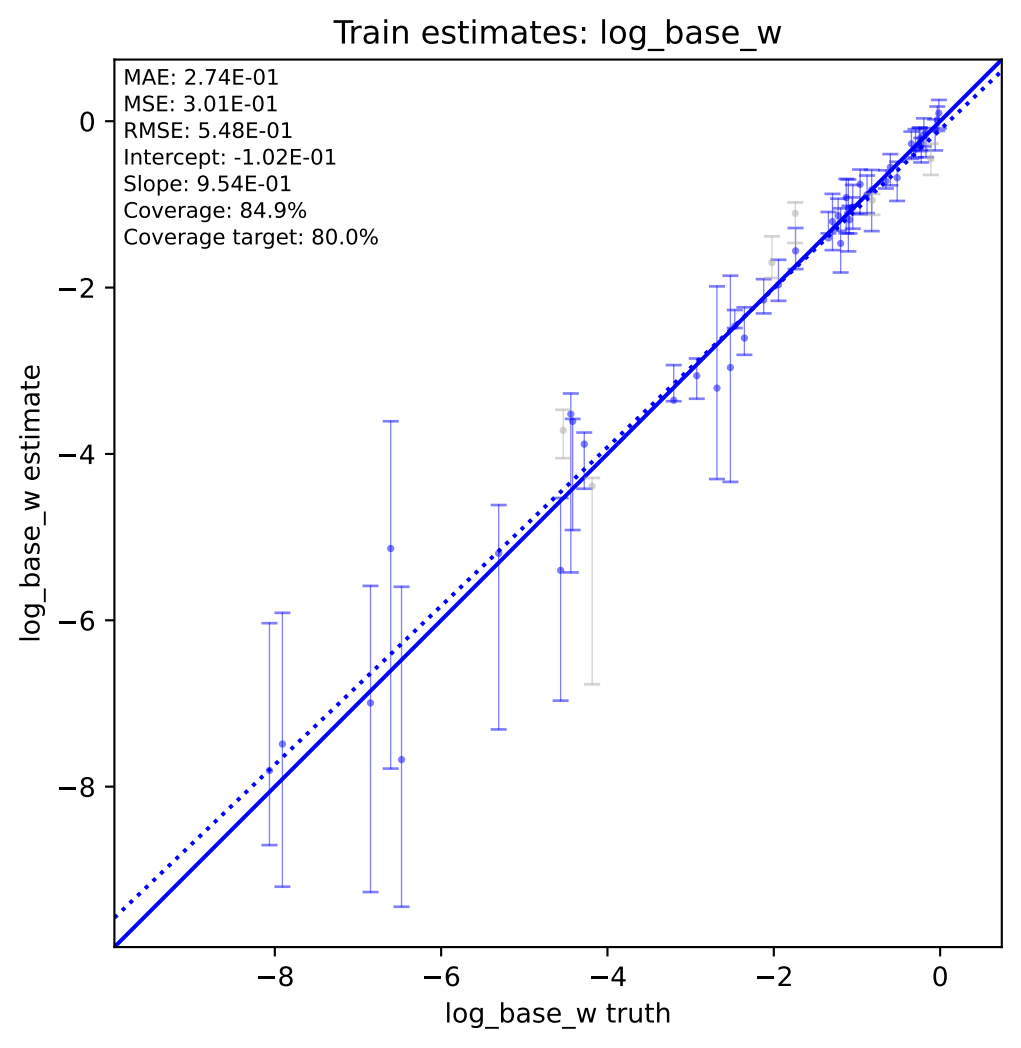

Density: training aux. data
N_bar
sim: 0.96 (0.69, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 0.18 (-0.88, 2.66)
emp: 2.53 (2.53, 2.53)
brlen_var
sim: -0.40 (-1.48, 2.25)
emp: 2.45 (2.45, 2.45)
colless
sim: -1.40 (-1.65, -0.67)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 3.66 (0.64, 8.81)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.04 (1.01, 4.50)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 2.00 (0.96, 2.19)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.09 (-0.42, 1.41)
emp: 1.52 (1.52, 1.52)
log10_brlen_mean
sim: -0.19 (-0.71, 1.17)
emp: 1.24 (1.24, 1.24)
log10_root_age
sim: 0.80 (0.25, 1.91)
emp: 1.88 (1.88, 1.88)
log10_tree_length
sim: 2.39 (1.77, 3.43)
emp: 3.11 (3.11, 3.11)
n_dat_0
sim: 6.47 (3.78, 7.48)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 5.89 (3.54, 6.84)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 187.00 (17.00, 284.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.27 (-0.36, -0.21)
emp: -0.83 (-0.83, -0.83)
Density
Variable

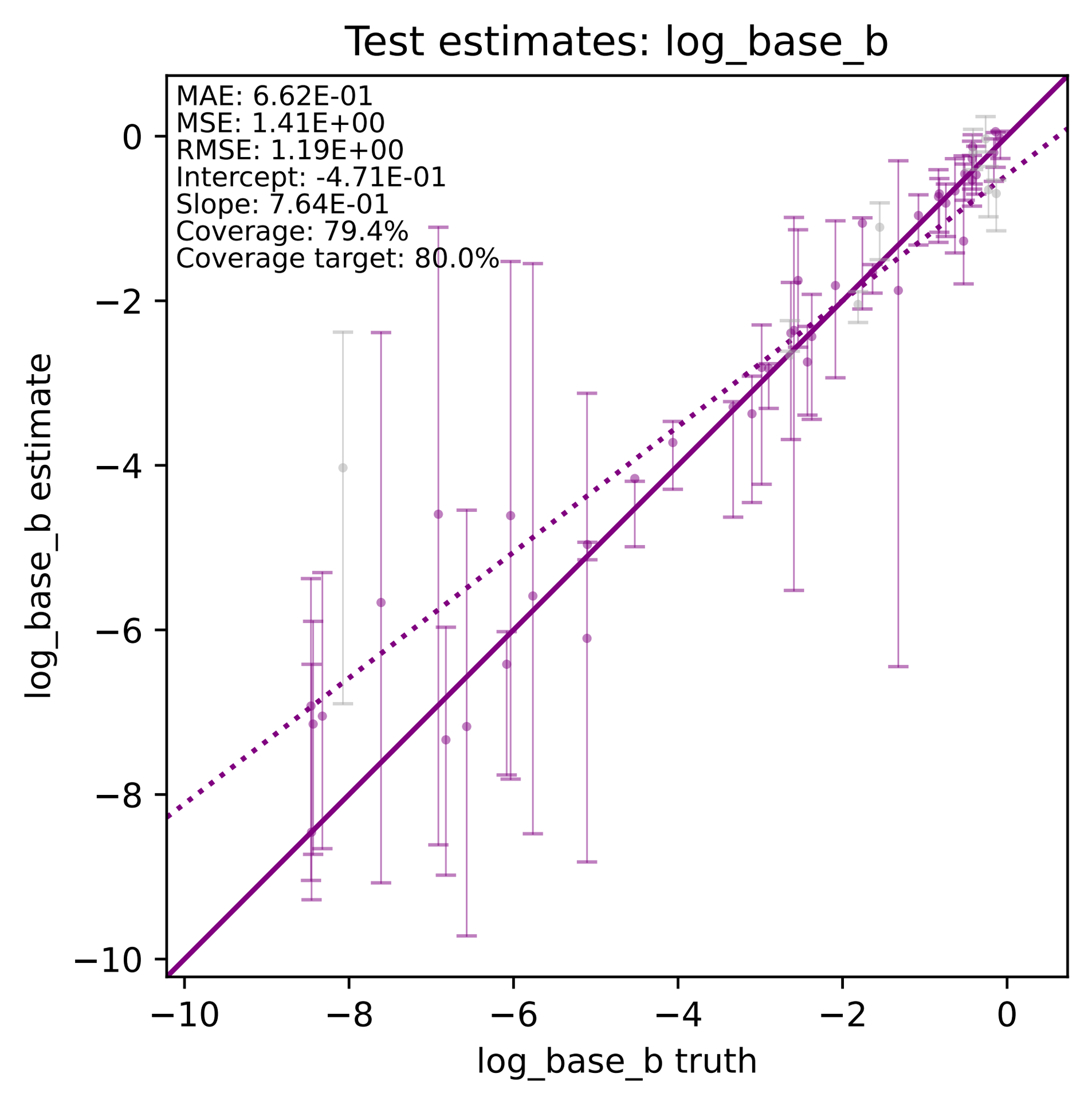

Density: training labels
log_base_b
sim: -1.60 (-8.59, -0.04)
emp: -7.60 (-7.60, -7.60)
log_base_d
sim: -1.24 (-8.42, -0.04)
emp: -9.45 (-9.45, -9.45)
log_base_e
sim: -2.36 (-8.85, -0.07)
emp: -6.57 (-6.57, -6.57)
log_base_w
sim: -1.28 (-8.28, -0.04)
emp: -3.59 (-3.59, -3.59)
−8
−6
−4
−2
0
Density
Variable

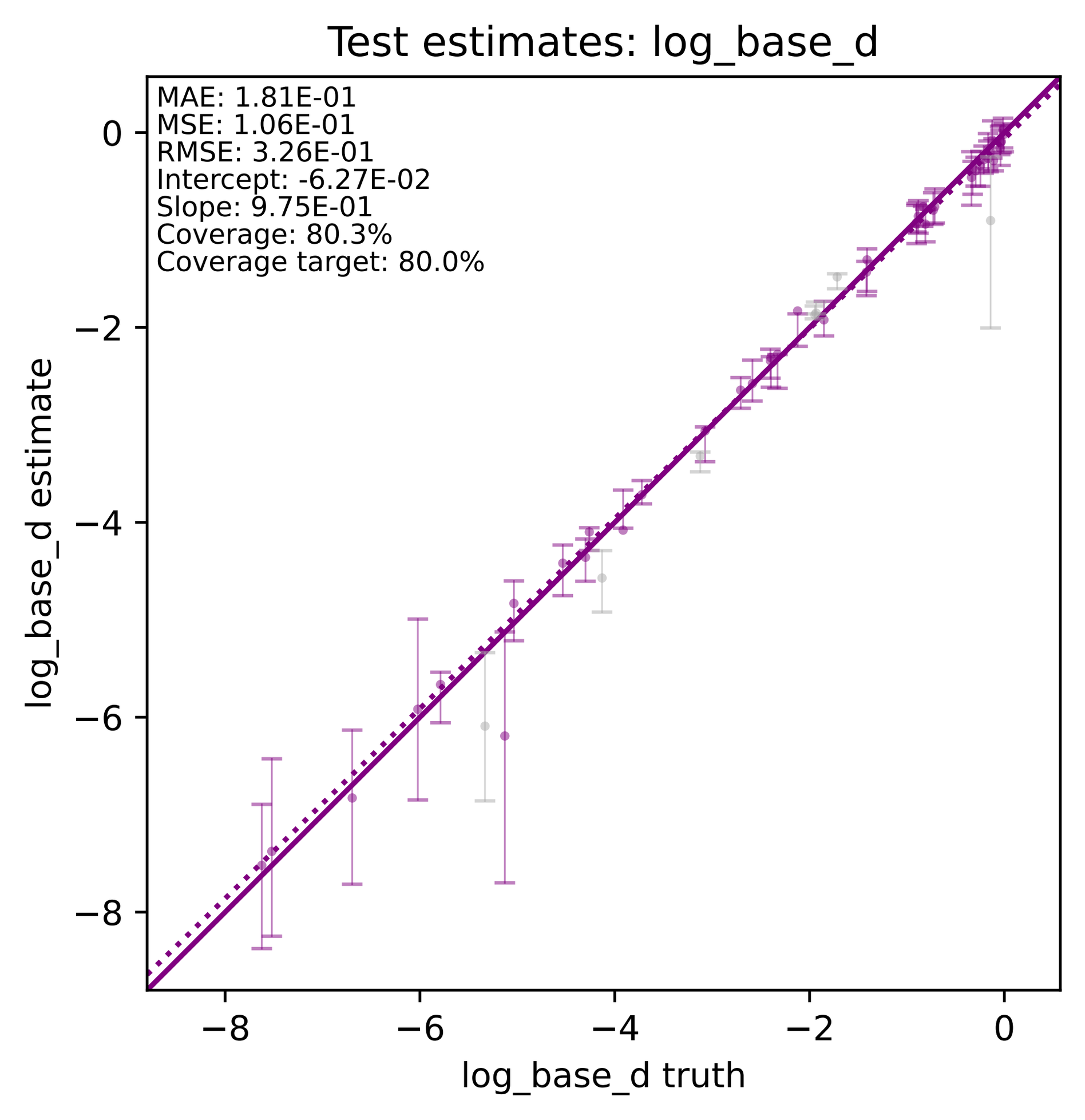

Train estimates: log_base_b
MAE: 6.03E-01
MSE: 1.16E+00
RMSE: 1.08E+00
Intercept: -3.90E-01
Slope: 8.06E-01
Coverage: 84.5%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

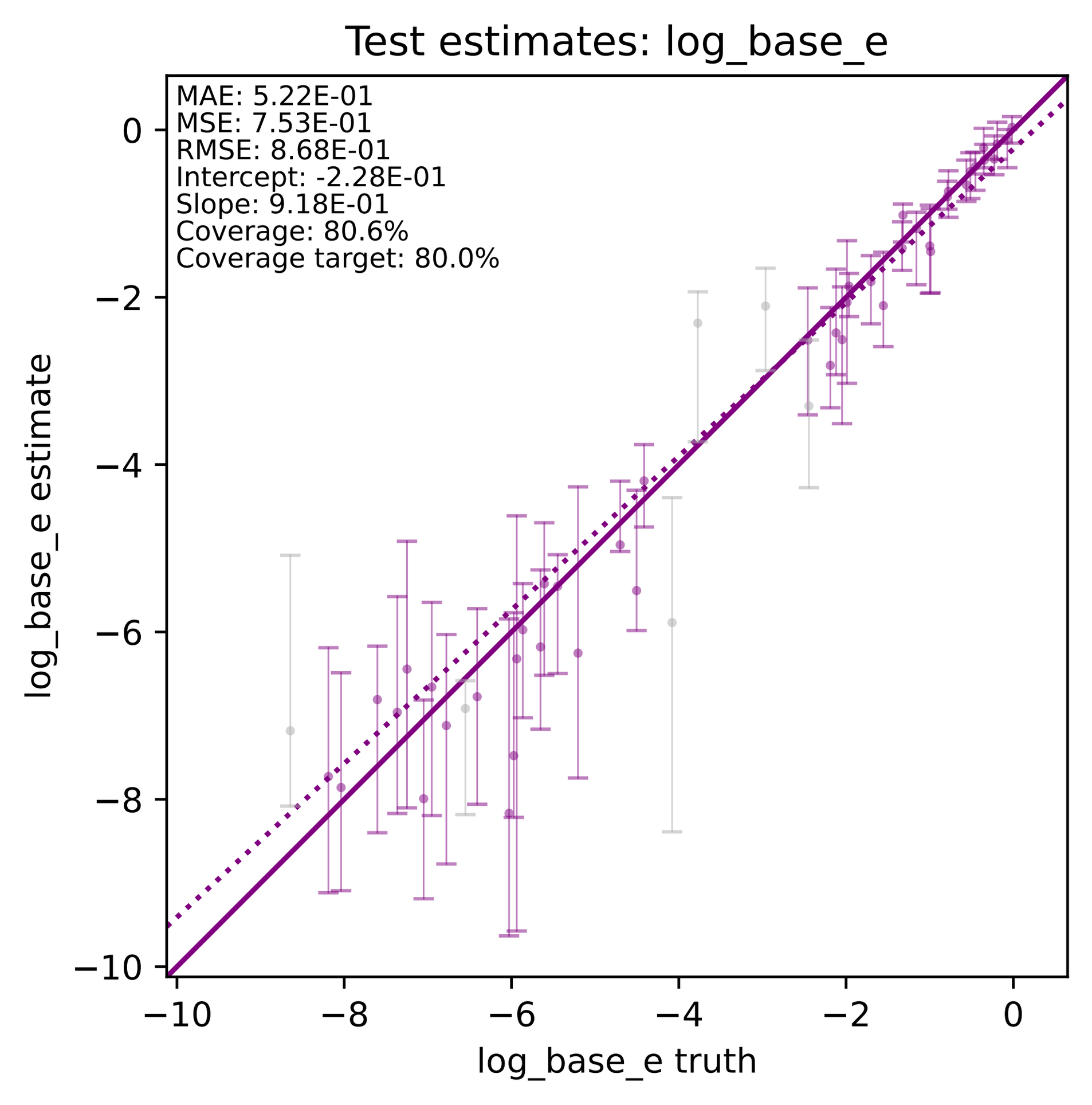

Train estimates: log_base_d
MAE: 1.73E-01
MSE: 1.07E-01
RMSE: 3.28E-01
Intercept: -8.04E-02
Slope: 9.71E-01
Coverage: 83.3%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

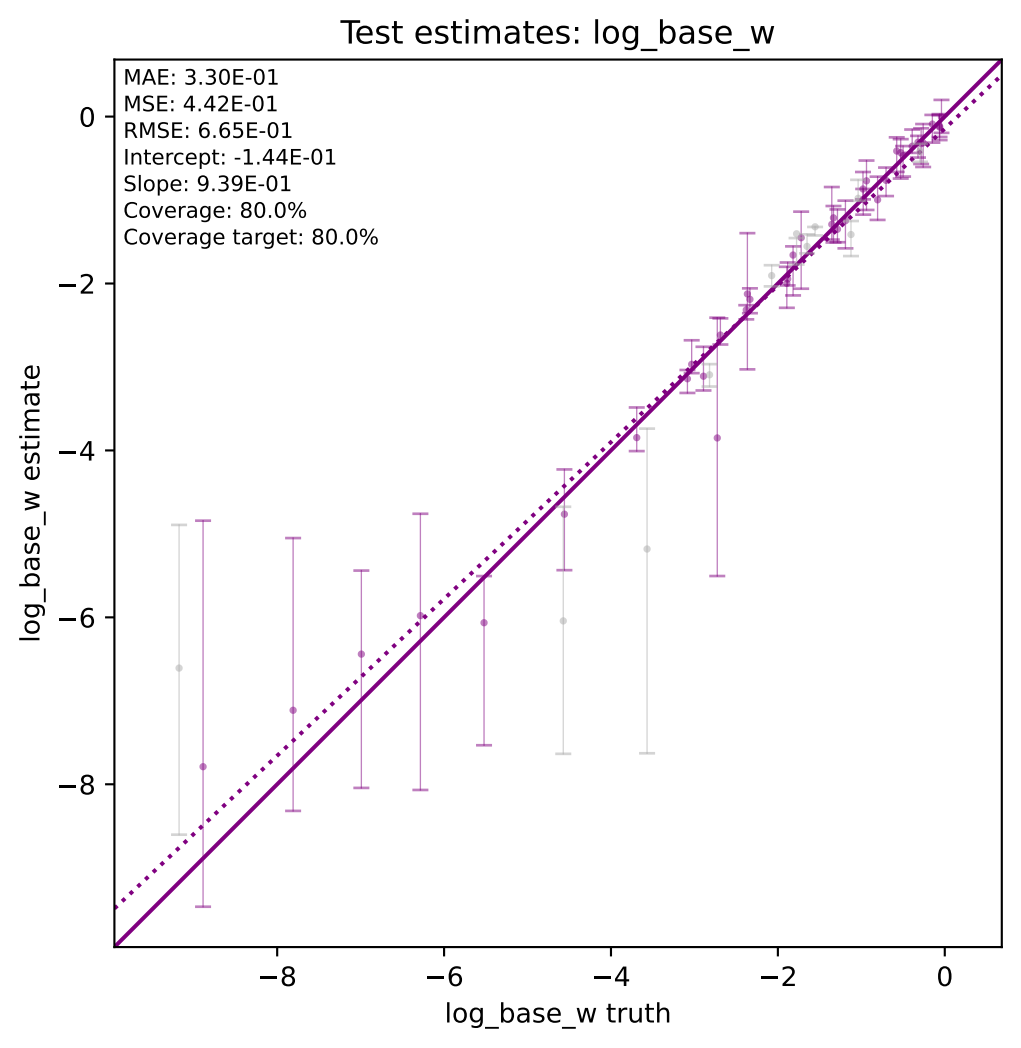

Train estimates: log_base_e
MAE: 4.53E-01
MSE: 6.01E-01
RMSE: 7.75E-01
Intercept: -2.18E-01
Slope: 9.18E-01
Coverage: 85.0%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

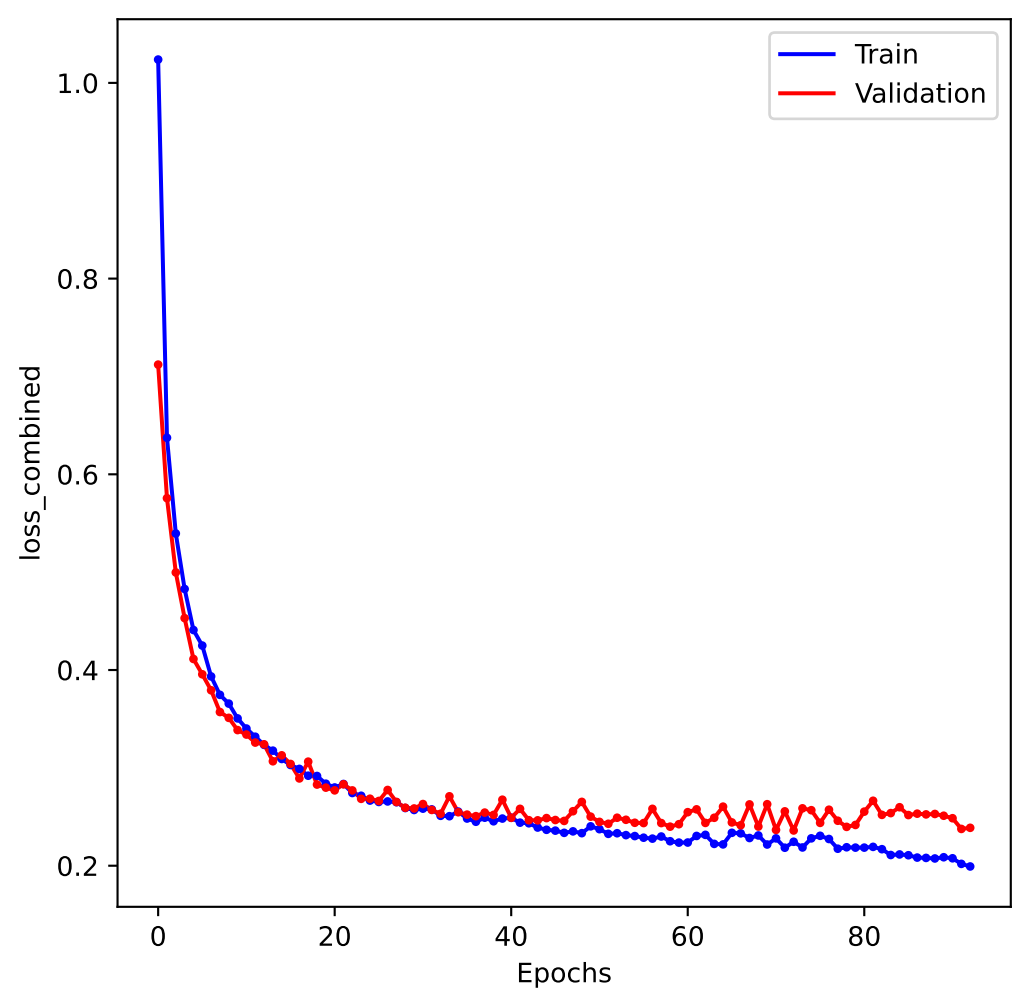

Train estimates: log_base_w
MAE: 2.74E-01
MSE: 3.01E-01
RMSE: 5.48E-01
Intercept: -1.02E-01
Slope: 9.54E-01
Coverage: 84.9%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

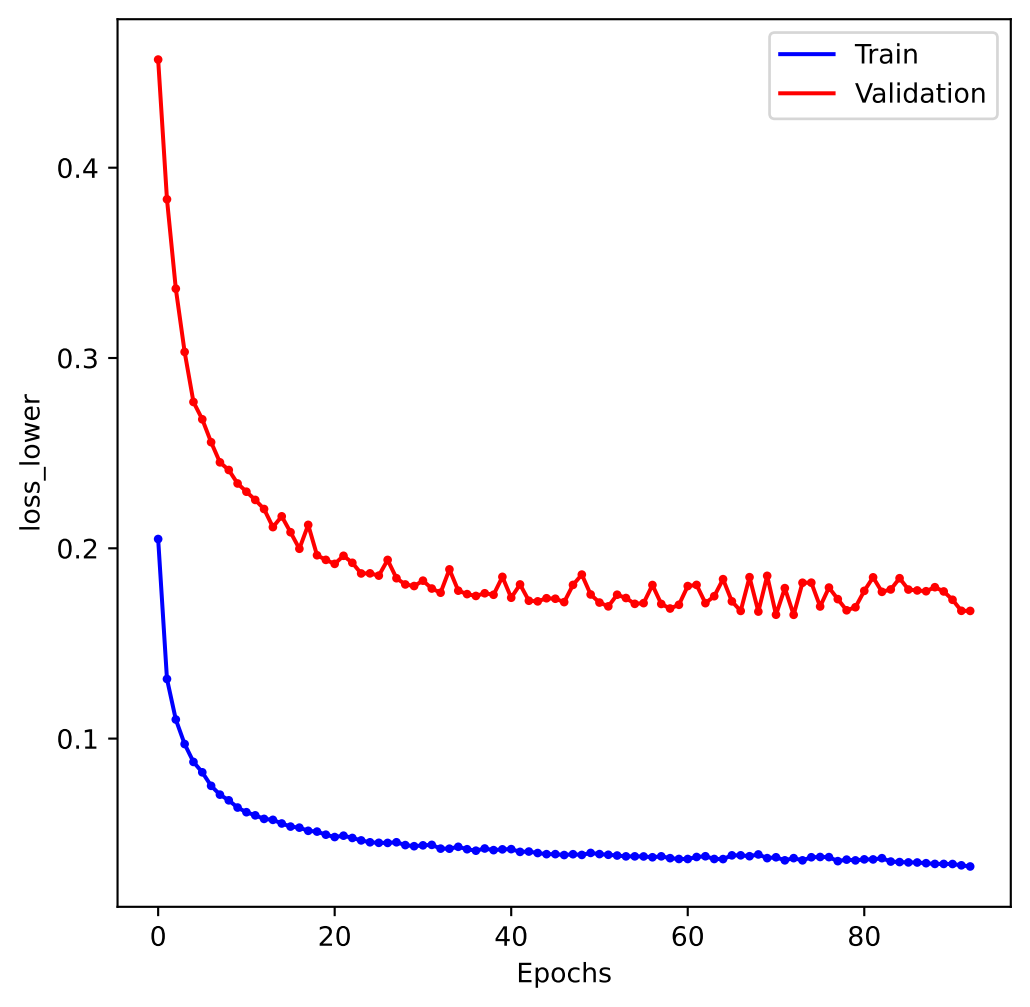

Test estimates: log_base_b
MAE: 6.62E-01
MSE: 1.41E+00
RMSE: 1.19E+00
Intercept: -4.71E-01
Slope: 7.64E-01
Coverage: 79.4%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

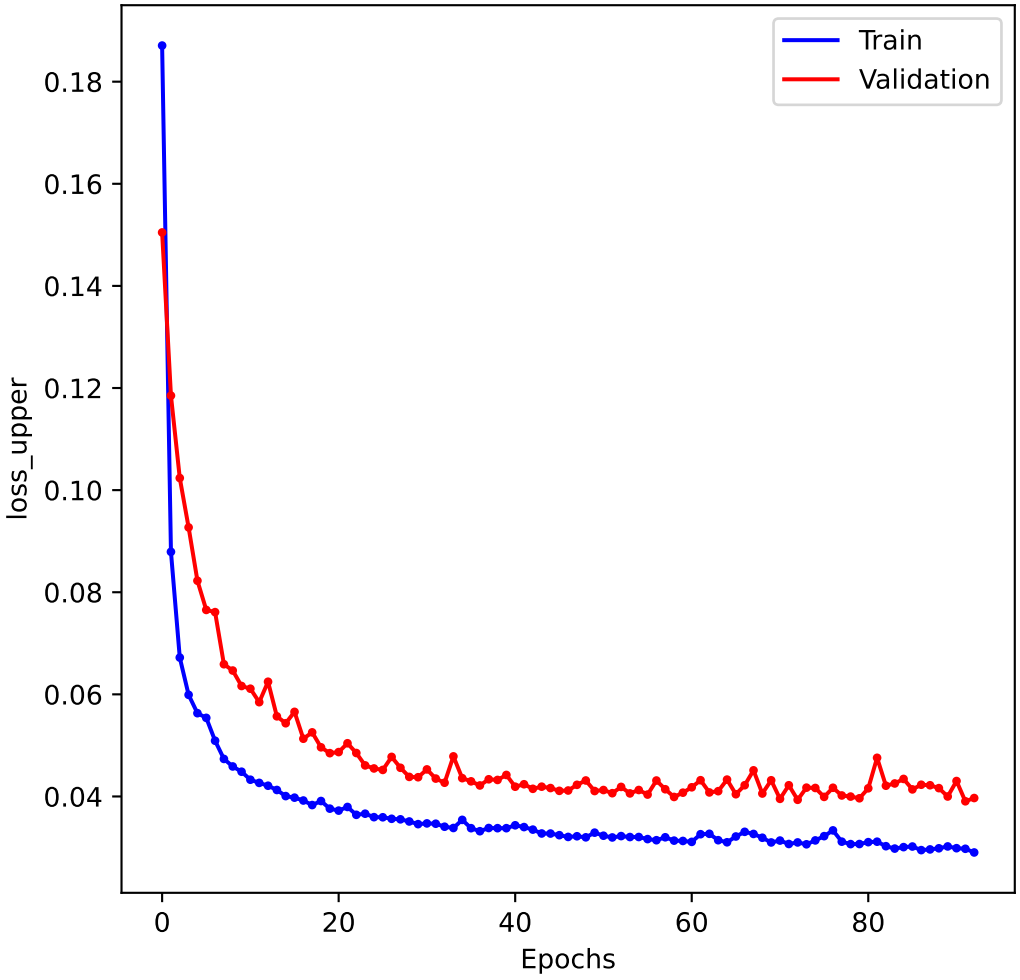

Test estimates: log_base_d
MAE: 1.81E-01
MSE: 1.06E-01
RMSE: 3.26E-01
Intercept: -6.27E-02
Slope: 9.75E-01
Coverage: 80.3%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

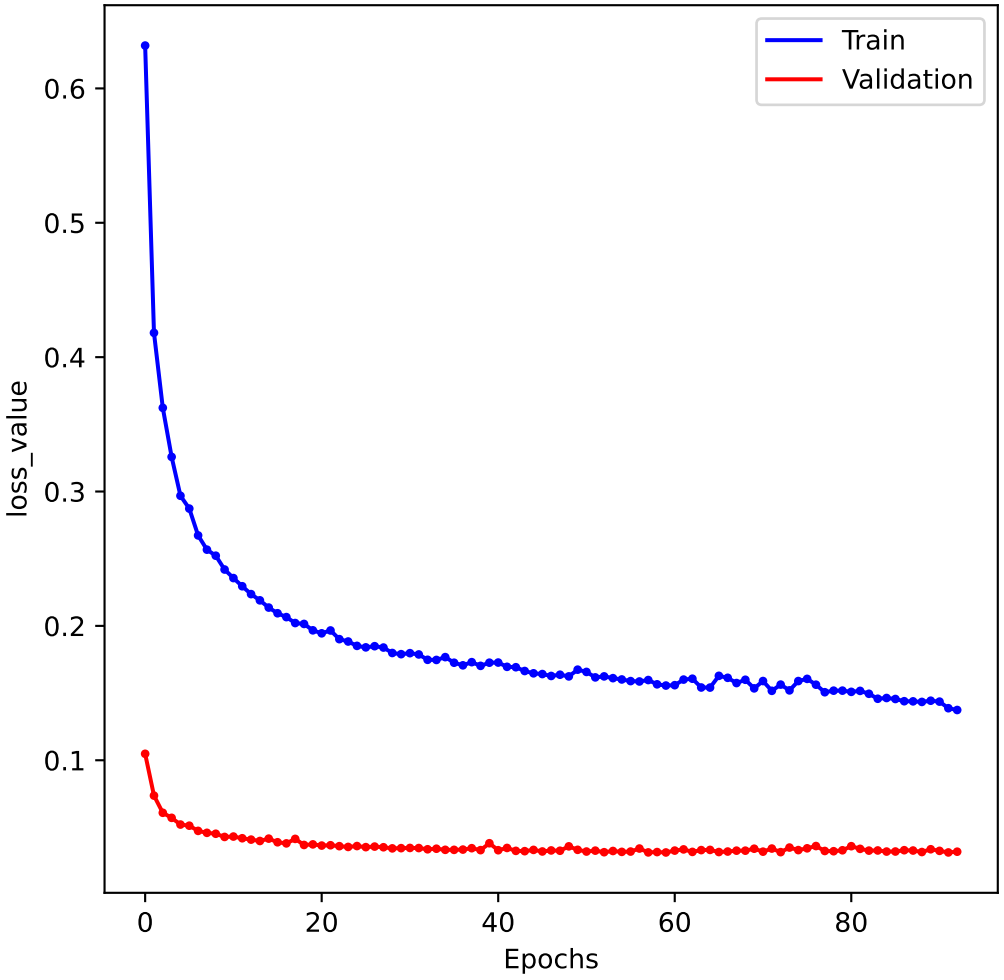

Test estimates: log_base_e
MAE: 5.22E-01
MSE: 7.53E-01
RMSE: 8.68E-01
Intercept: -2.28E-01
Slope: 9.18E-01
Coverage: 80.6%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

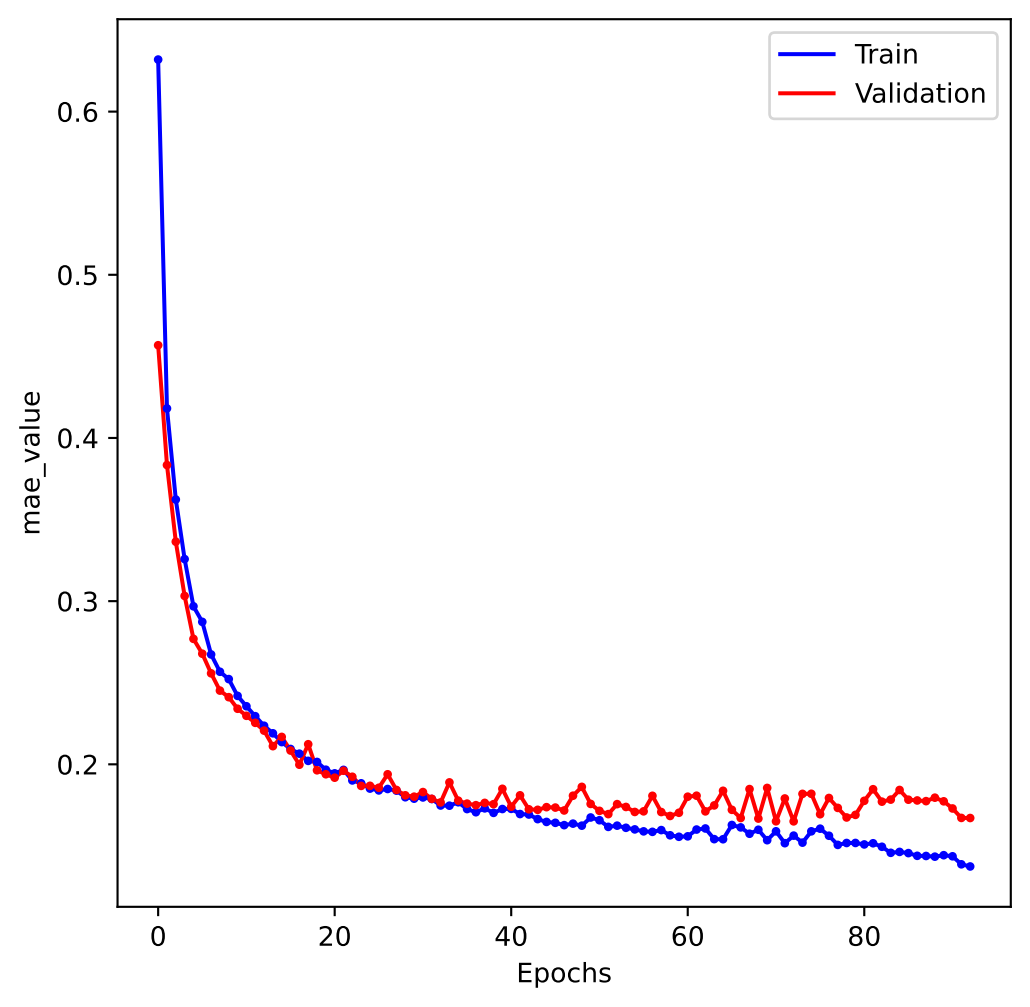

Test estimates: log_base_w
MAE: 3.30E-01
MSE: 4.42E-01
RMSE: 6.65E-01
Intercept: -1.44E-01
Slope: 9.39E-01
Coverage: 80.0%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

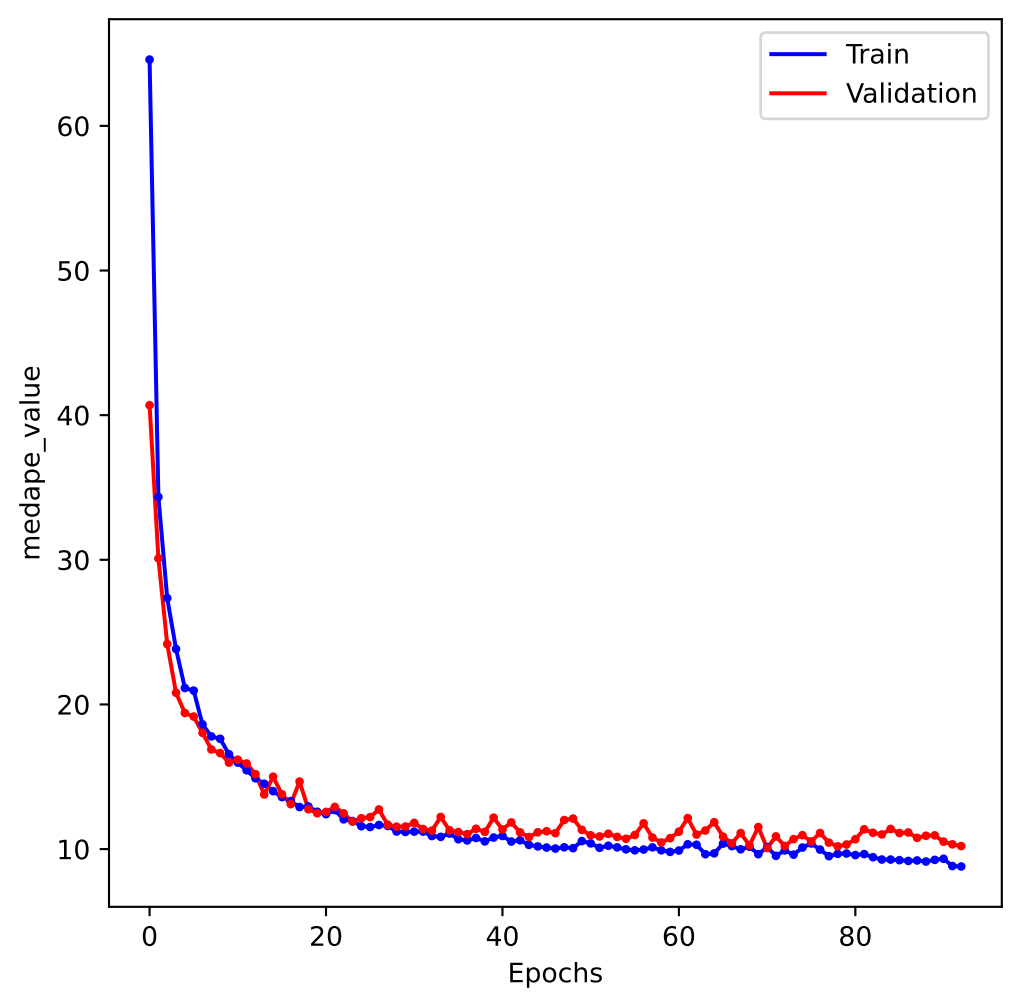

Train
Validation
loss_combined
1.0
0.8
0.6
0.4
0.2
0
20
40
60
80
Epochs

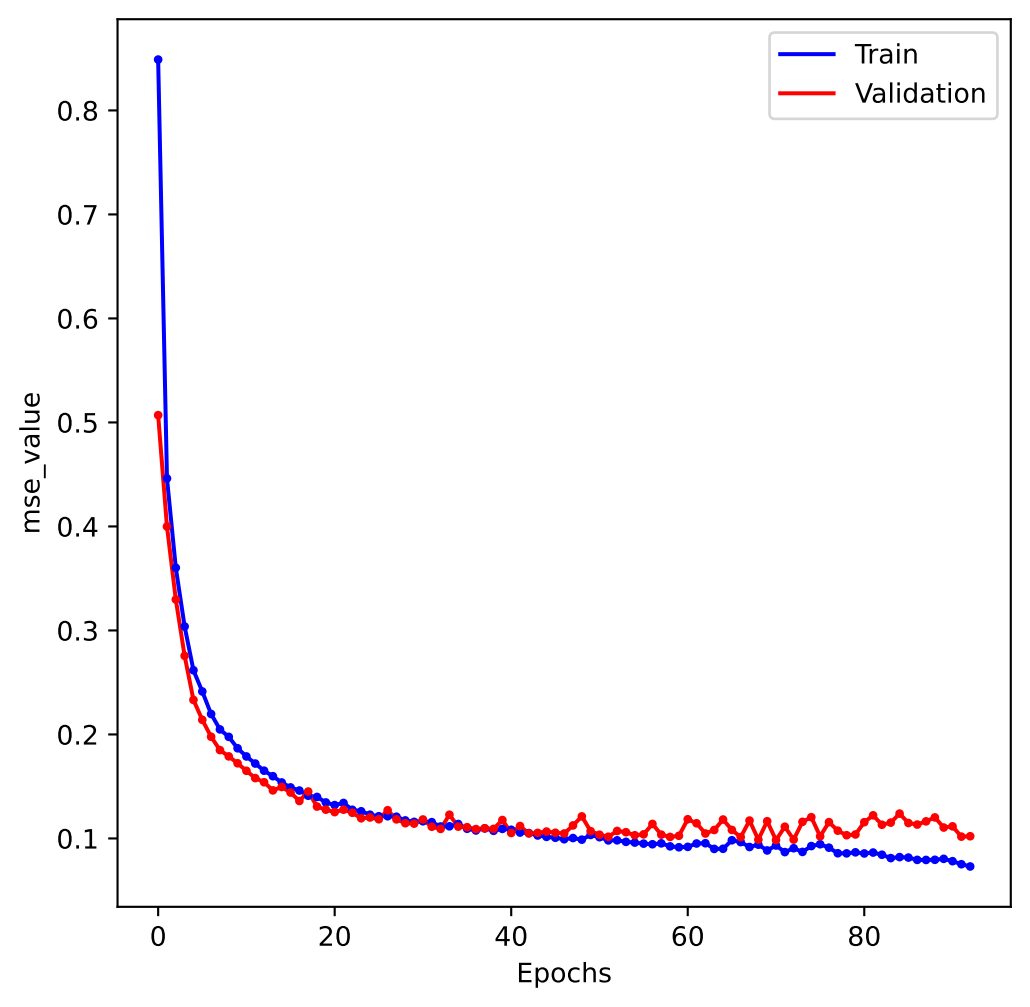

Train
Validation
0.4
0.3
0.2
0.1
loss_lower
0
20
40
60
80
Epochs

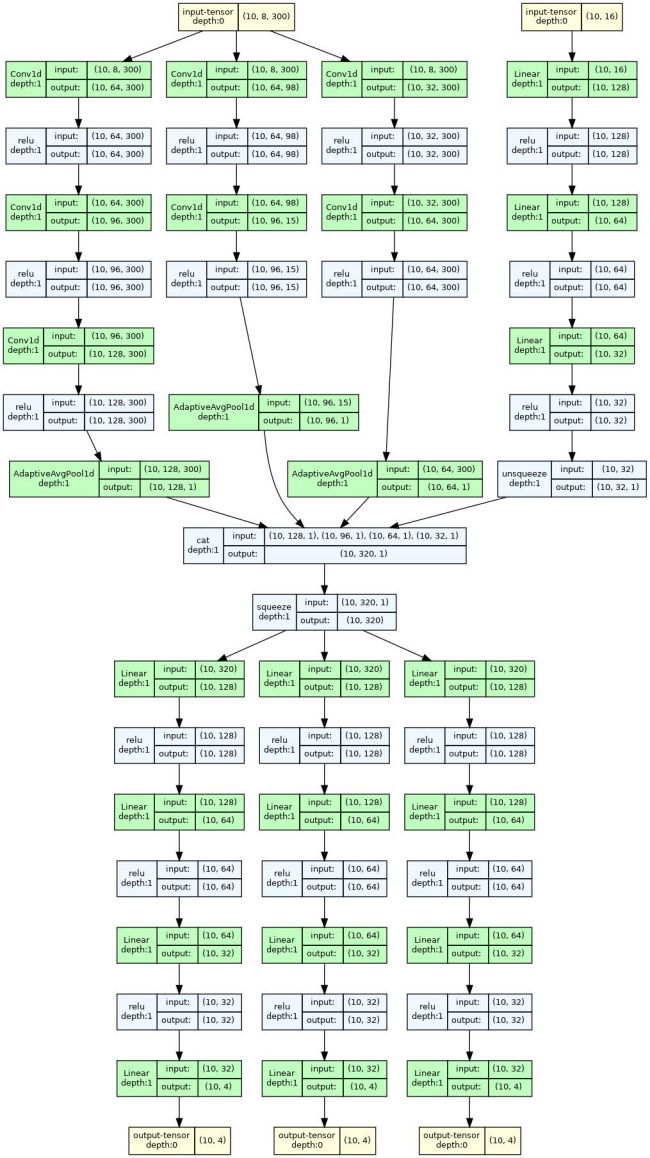

Train
Validation
loss_upper
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0
20
40
60
80
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
Epochs


Train
Validation
mae_value
0.6
0.5
0.4
0.3
0.2
0
20
40
60
80
Epochs


Train
Validation
medape_value
60
50
40
30
20
10
0
20
40
60
80
Epochs


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 4)
Linear depth:1
input: (10, 32)
output: (10, 4)
Linear depth:1
input: (10, 32)
output: (10, 4)
output-tensor depth:0
(10, 4)
output-tensor depth:0
(10, 4)
output-tensor depth:0
(10, 4)